TSM 099

Homework 3

Common factors and advanced factoring

Topics 13-17 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Work without a calculator before checking. Give exact, simplified answers.

GCF first

1. Find the GCD of 72 and 108. Then find the greatest common monomial factor of 18x cubed y and 30x squared y cubed.

2. Factor out the GCF.

$18x^3y - 30x^2y^2$

Factor trinomials

3. Factor completely over the integers. Verify by multiplication.

$x^2 - 2x - 24$

4. Factor completely over the integers. Verify by multiplication.

$x^2 - 11x + 24$

5. Factor completely over the integers. Verify by multiplication.

$6x^2 + 7x + 2$

6. Factor completely over the integers. Verify by multiplication.

$8x^2 - 2x - 3$

7. Factor completely over the integers. Verify by multiplication.

$10x^2 - 13x + 3$

8. Factor completely over the integers. Verify by multiplication.

$12x^2 + 11x + 2$

Grouping and complete factorization

9. Factor by grouping.

$x^3 + 2x^2 - 9x - 18$

10. Factor by grouping.

$6x^3 - 9x^2 + 4x - 6$

11. Factor completely.

$3x^3 - 12x$

12. Factor completely.

$2x^2 - 8x + 8$

13. Factor completely.

$9x^2 - 25$

14. A student factors 4x squared - 16 as 4(x squared - 4) and stops. Explain why the answer is incomplete and finish it.